Track 2 report — which approved medicines could act on this child's broken checkpoint?

Standing: draft, written 2026-09-18 from executed runs (iteration 2, step 7 — methodology entries 110–113). Every number is read from a table with a provenance manifest, every claim was registered before its test, and every cited paper was resolved against PubMed the same day. **Superseded by `../iteration-2/` and never uploaded** — kept for provenance, with its facts frozen at mechanism card **version 3** (the aneuploidy burden, fact F8, is *unmeasured* here). Restructured on 2026-09-20 to the same four-criteria shape as iteration 2, **with every fact unchanged at card v3**: sections were added and renumbered, nothing was remeasured and no later result was imported.

Written for the panel, in plain words. Each technical term appears once with its meaning in brackets; the full dictionary is served at `/learn` on the dashboard.

> The challenge's own framing, kept in view throughout: *submissions are hypotheses for follow-up, not evidence that a medicine works.* Nothing below is a recommendation to treat.

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1. How this report answers the panel's four criteria

> **Rubric:** all four — this section is the map, and the last column names each criterion's weakest point before a judge has to go looking for it.

criterion	weight	where in this report	the evidence, in one line	the weakest point we know
Scientific Rigor	35 %	§2, §4, §5, §6, §10, §12	the mechanism is read off a genotype the challenge's answer key confirmed, with nine facts measured onto the card; every approved drug sits on a five-rung ladder written before the search; six claims were registered with their refutation conditions before any drug-level query	**no approved drug has a molecule-level result on BubR1 itself.** The deepest mechanism lead (D3) is one mouse paper, and the two approved molecules inherit it as a *proxy*
Potential Impact	25 %	§7, and the do-not table in §6.3	a do-not table naming 13 spindle poisons — two of them this child's standard chemotherapy — that work by relying on the very checkpoint that is broken; the D1 result and that caution follow from the checkpoint, so they are arguments about the disease, not only about this gene	the child's **aneuploid fraction is unmeasured in this draft**, so direction D2 is argued from published aneuploid-cell biology and not from him; and nothing here is evidence that a medicine works
Innovation	25 %	§8, §5.3	one structured mechanism card drives every query; the same pipeline is re-run on a decoy gene as a negative control and the overlap is measured against a threshold fixed in advance; automated and person-read evidence are printed side by side; the do-not table is a first-class output	these are **process** innovations, not new biology: no new model, no new algorithm, and knowledge-graph repurposing rankers already exist (§8)
Scalability	15 %	§9, §11	one card in, five commands out, every table carrying its own provenance manifest; the whole run is two laptop runs of about ten minutes against public APIs, seconds when cached, no cloud	the curated rungs are the human bottleneck — 13 abstracts read by one person on one day — and the transfer run onto a genuinely different case **post-dates this draft** and belongs to iteration 2 (§9)

2. The answer in one page

> **Rubric:** Scientific Rigor — the whole claim on one page: the mechanism read off a confirmed genotype, the four directions with the highest rung each reached, and what is not known. Every later section exists to support or qualify something stated here.

The child. A boy with mosaic variegated aneuploidy (MVA: in some of his cells, a chromosome is missing or duplicated, and different cells are wrong in different ways), short stature, failure to thrive, calcium deposits in the kidneys, and a rhabdomyosarcoma (a muscle-tissue cancer of childhood). Eight clinical terms were delivered with the case.

The cause. Track 1 found, and the challenge's answer key confirmed, two different broken copies of one gene, `BUB1B` — one from each parent (a *compound heterozygote*). The gene makes BubR1, the protein that acts as the **brake** on cell division: it holds a dividing cell until every chromosome is properly attached before the cell is allowed to split. One copy makes no protein at all (a *stop* change); the other makes a weakened protein (a *missense* change in the protein's last stretch). The child therefore has **some brake, but less than normal** — which is exactly the pattern every published MVA case with this gene shows, because having no brake at all is not survivable.

The mechanism, in the form the methods form asks for. *Loss of function*, not gain: less BubR1, a leakier checkpoint, chromosomes that separate too early, cells with the wrong count, and — in surviving cells — a chronic stress from carrying the wrong number of copies of many proteins. Downstream: the aneuploidy that names the disease, the growth problems, and a predisposition to exactly the cancers this child has.

Four ways a drug could act, graded on a ladder written before the search.

Direction	The idea	What we found	Best rung
D1 reinforce the brake	make the remaining checkpoint stronger	**no approved drug does this.** Every checkpoint drug on the market or in trials *weakens* it. D1 became the do-not table	—
D2 exploit the stress of aneuploid cells	aneuploid cells are already stressed; some approved drugs tip them over	**chloroquine** (a malaria drug) killed aneuploid cells preferentially in the reference screen; metformin, hydroxychloroquine and bortezomib inherit the result through their drug class	2 (cell model)
D3 raise the amount of working BubR1	the weakened copy still works; stabilise it	**niacin / nicotinamide** (vitamin B3 forms): boosting the NAD^+ pathway raised BubR1 in mice bred to have too little of it; valproic acid and vorinostat act on BubR1's acetylation switch	3 (animal model)
D4 the child's actual tumour	approved, non-chemotherapy drugs with a mechanism in rhabdomyosarcoma	**statins** (lovastatin, simvastatin, atorvastatin): rhabdomyosarcoma cells depend on making their own cholesterol; lovastatin itself slowed them; simvastatin has already been in a completed paediatric sarcoma trial	5 (approved, trialled in this tumour, used in children) — carried by rung-2 mechanism evidence

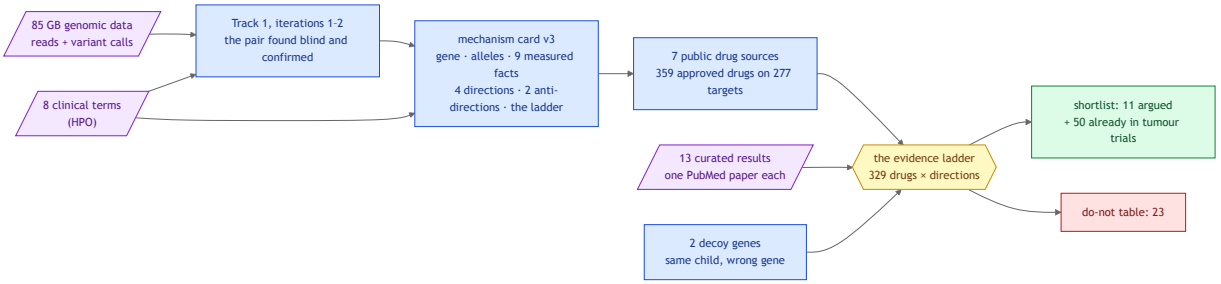
The half a clinician may care about most: what not to do. Thirteen approved drugs are *spindle poisons* — they work by jamming cell division and relying on the checkpoint to hold the cell. In a child whose checkpoint is the broken part, that reliance is a hypothesis worth weighing before treatment. Several of them (vincristine, vinorelbine) are standard rhabdomyosarcoma chemotherapy. **This report flags the question for the treating team; it does not advise withholding standard care.**

What we do not know. How aneuploid this child's cells actually are (the sequencing data delivered cannot measure it with any power; the analysis that could is planned and not run). Whether the weakened BubR1 copy is unstable or simply inactive. And whether any drug on this page helps — rung 3 is one mouse paper.

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3. Where this comes from — two datasets, one line of evidence

> **Rubric:** Scientific Rigor — shows that both delivered datasets (the DNA and the clinical terms) enter the argument, and where each stage's output can be checked.



> **Legend — purple = raw inputs** (the child's DNA, the child's clinical terms, and the papers a person read); **blue = machinery** (every box is a tool with a manifest); **yellow = the decision point** (the ladder); **green = the goal** (the shortlist); **red = what to avoid** (the do-not table).

Both datasets are used at every stage: the genomic data fixed the gene and the two alleles; the clinical terms fixed the disease, the tumour and — through the tumour — the fourth direction and the paediatric questions. Where a result uses only one, the table says so.

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4. The mechanism (methods form Q14: loss- versus gain-of-function, pathway, consequence)

> **Rubric:** Scientific Rigor — this is the methods form's Q14, the variant-mechanism characterisation. Every drug argument below is only as sound as this section.

4.1 Loss of function, of a specific shape

copy	what changed	what it does to the protein	evidence
one	a *stop* change in exon 17 of 23	the protein ends early, before its last working part — **no functional protein**	catalogued as disease-causing for MVA (ClinVar VCV000533901)
the other	a *missense* change in exon 23 of 23 — one amino acid swapped	a **weakened** protein: never seen in 140,000 healthy genomes, predicted damaging by two independent predictors	Track 1 (iteration 1, entries 070–074; iteration 2, entry 099)

Nine facts measured in iteration 2 sit on the mechanism card so the argument does not rest on the allele labels alone:

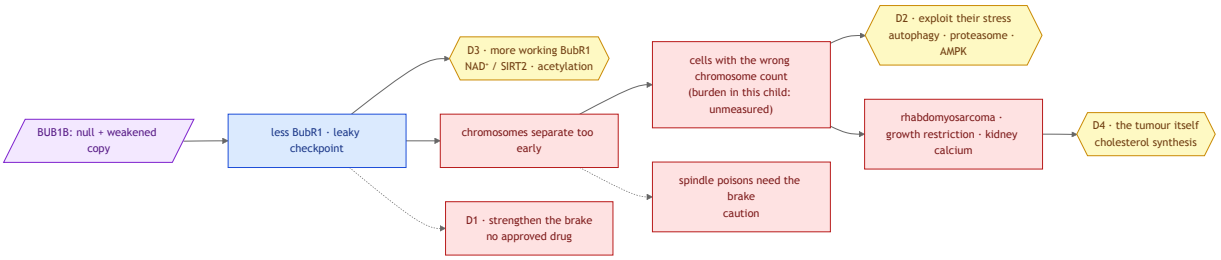
fact	measured	what it changes for a drug
the two copies are on **different** chromosomes (*trans*)	**confirmed** by the challenge's answer key (a full match at rank 1)	the whole argument stands on a confirmed genotype, not an inference
each copy present **once** in the raw sequencing reads	yes — both alleles at one-copy dosage, read without an aligner	a *null + weakened* genotype: some BubR1 exists to stabilise (D3 is live)
no deleted or duplicated stretch in the gene	every exon at two copies; no structural variant in the gene	the two alleles are the whole story
where the missense sits in the protein	outside every catalogued domain, in a region the structure model is confident about	we **cannot** say whether the weakened copy is unstable or inactive; D3 is argued as *more BubR1*, not *refold it*
how aneuploid this child's cells are	**unmeasured** — the karyotype test iteration 1 ran has no power for mosaic gains; the powered analysis (step 8) has not run	D2 is argued from published aneuploid-cell biology, not from this child's burden

(Complete list F1–F9, with the entry each comes from: mechanism card v3, and step 7's README.)

4.2 The pathway: the spindle assembly checkpoint

BubR1 is a core part of the **spindle assembly checkpoint** (SAC — the wait-until-attached brake) and of the complex that switches off the machinery that pulls chromosomes apart. Less BubR1 means a **leaky brake**: division can proceed with chromosomes still unattached. Published MVA patient cells show strongly reduced BubR1 and a weakened checkpoint (Suijkerbuijk 2010, PMID 20516114); the founding paper tied biallelic `BUB1B` changes to MVA with cancer predisposition (Hanks 2004, PMID 15475955). Complete loss is lethal before birth in mouse and human — which is why "switch BubR1 off" is never a therapeutic direction here.

4.3 Downstream consequences, and where a drug could stand



> **Legend** — **purple** the genotype; **blue** the molecular consequence; **yellow** the three places an approved drug could stand, each a direction below; **red** what goes wrong, and the two dotted boxes are the direction with no drug and the class to be cautious with.

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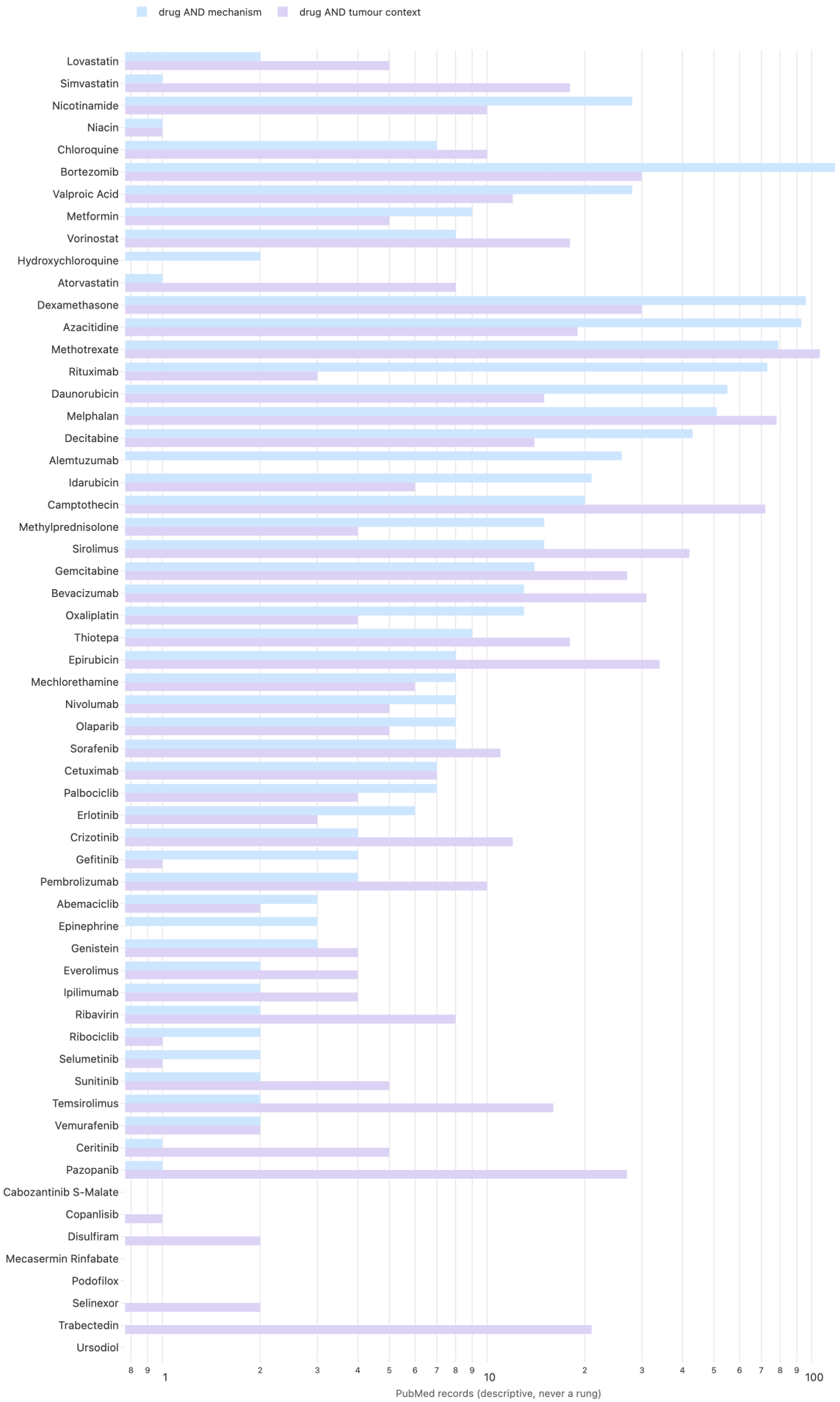
5. How the candidates were found and graded

> **Rubric:** Scientific Rigor, and the negative control that §8 claims as the new part — how a candidate got in, how it was graded, and what the pipeline returns for a gene that should return nothing.

5.1 The search: one card, seven public sources, no hand-typed query

The mechanism card (a structured file naming the gene, the two allele classes, the pathway, the disease, the tumour and four *directions* with seed genes) drives every query. The Reactome pathway map widens the seed genes into 277 targets; Open Targets, ChEMBL and DGIdb list every drug known to act on them (3,859 drug–target rows, **359 market-approved drugs**); openFDA confirms a current US label; ClinicalTrials.gov and PubMed answer the trial and literature questions. Every table carries a manifest with the retrieval time and a digest of every reply, and the same command re-runs it for any other case by swapping the card.

One of those per-drug queries is worth showing, because it is the one a reader is most likely to mistake for evidence:


drug AND mechanism
drug AND tumour context
Lovastatin
Simvastatin
Nicotinamide
Niacin
Chloroquine
Bortezomib
Valproic Acid
Metformin
Vorinostat
Hydroxychloroquine
Atorvastatin
Dexamethasone
Azacitidine
Methotrexate
Rituximab
Daunorubicin
Melphalan
Decitabine
Alemtuzumab
Idarubicin
Camptothecin
Methylprednisolone
Sirolimus
Gemcitabine
Bevacizumab
Oxaliplatin
Thiotepa
Epirubicin
Mechlorethamine
Nivolumab
Olaparib
Sorafenib
Cetuximab
Palbociclib
Erlotinib
Crizotinib
Gefitinib
Pembrolizumab
Abemaciclib
Epinephrine
Genistein
Everolimus
Ipilimumab
Ribavirin
Ribociclib
Selumetinib
Sunitinib
Temsirolimus
Vemurafenib
Ceritinib
Pazopanib
Cabozantinib S-Malate
Copanlisib
Disulfiram
Mecasermin Rinfabate
Podofilox
Selinexor
Trabectedin
Ursodiol
8 9 1 2 3 4 5 6 7 8 9 10 2 3 4 5 6 7 8 9 100
PubMed records (descriptive, never a rung)

> **Legend** — one row per shortlisted drug, the eleven argued drugs first, then the fifty already in rhabdomyosarcoma trials. The axis is PubMed record counts on a **log scale**, so equal bar lengths mean equal *ratios*, not equal counts. **Blue** = records matching the drug *and* this mechanism (the gene, the protein, the pathway or the disease); **purple** = records matching the drug *and* the tumour context. A missing bar is a count of zero, not missing data. These counts are **descriptive and never move a rung**: the longest bars belong to the most-written-about drugs (bortezomib, dexamethasone, methotrexate), which is a fact about the literature, not about the strength of the argument for them here.

5.2 The ladder: written before the search

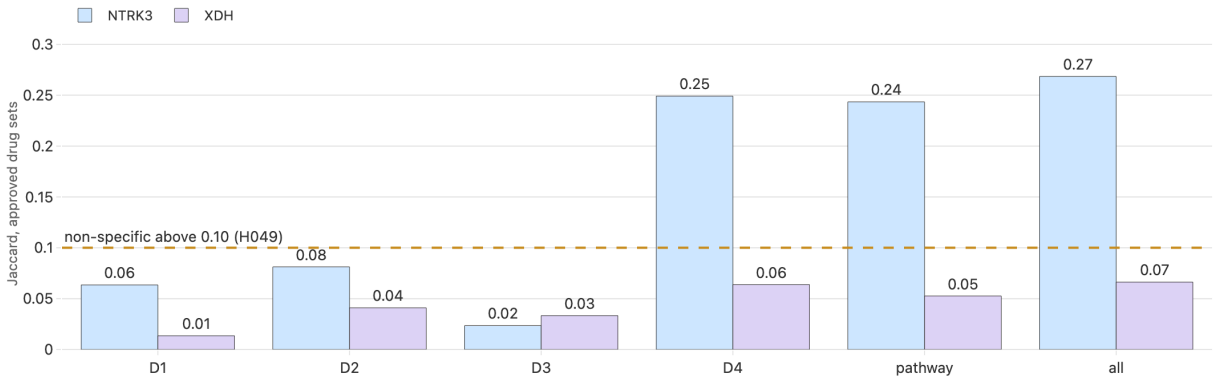
Every approved drug, on every direction it touches, is placed on **the highest rung it passes**:

rung	meaning	how it is tested
5	approved for a related indication, with paediatric use	approved **and** a registered trial in the child's tumour **and** a paediatric pointer (a label section 8.4, or a registered study admitting children)
4	clinical trial in a related indication	a registered trial in the child's tumour
3	animal model of the mechanism	a paper, read by a person, with its PubMed id
2	cell model of the mechanism	a paper, read by a person, with its PubMed id
1	mechanism only	the drug acts on a target in the direction's set — *named, never submitted*

Two rules the ladder prints rather than hides. **Automated and curated tests sit side by side:** rungs 4–5 are looked up by a program and say "already being tried in this tumour"; rungs 2–3 come from a paper a person read, and the row carries the PubMed id. **Inherited evidence is marked:** when the paper tested a same-axis compound and not the approved drug (the screen used AICAR; the approved drug is metformin), the row says `proxy` and names what was tested.

5.3 A control the search had to pass: the same child, a different gene

Track 1 tested every ranker against decoy genes; Track 2 does the same. The mechanism card was copied with the child's clinical side kept and the gene swapped — once for **NTRK3**, drawn by a seeded rule from Track 1's matched decoy set, once for **XDH**, a metabolic enzyme with nothing to do with cell division — and the unchanged pipeline ran on both. Whatever came out for the decoy as readily as for `BUB1B` was nominated by the clinical queries or by database density, not by the mechanism.



> **Legend** — one bar per decoy gene; the dashed yellow line is the pre-registered 0.10 threshold. **D4 (the tumour direction) is above it** against NTRK3 (Jaccard 0.25): the cancer pharmacopoeia comes out for any cancer-adjacent gene. **D1–D3 are below it** against both decoys: the mechanism directions are specific to the gene. The

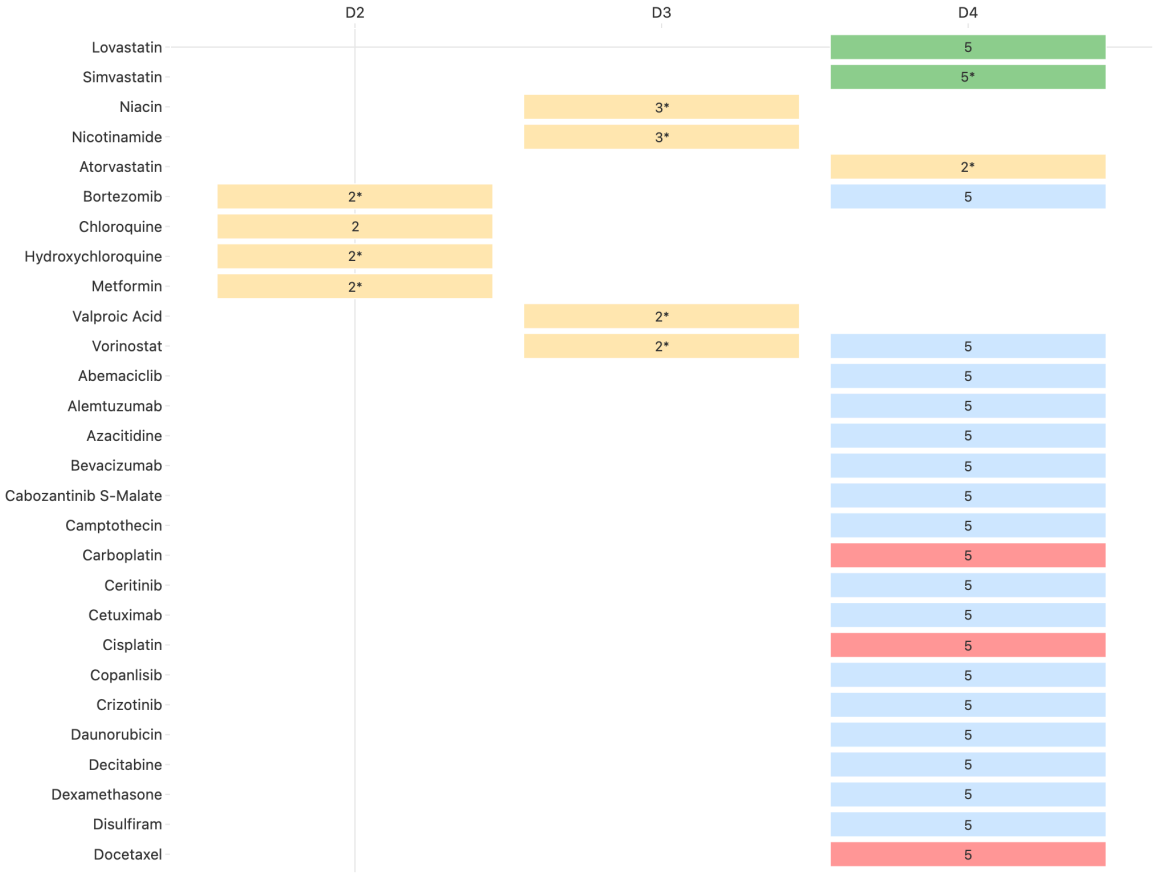
> consequence, applied: D4 is argued only from curated mechanism results, never from "this drug has a rhabdomyosarcoma trial".

The control also caught a flaw in the first grading run: a rhabdomyosarcoma trial was lifting drugs onto rung 5 on *mechanism* directions (85 drugs, tetracycline among them). A trial in the tumour says nothing about a checkpoint; the rule was narrowed to the tumour direction and the run repeated before any verdict — the change is recorded with before/after counts in entry 113.

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6. The candidates

> **Rubric:** Scientific Rigor — whether each proposed drug is actually supported by the mechanism, drug by drug, rung by rung, with the inherited evidence marked as inherited.



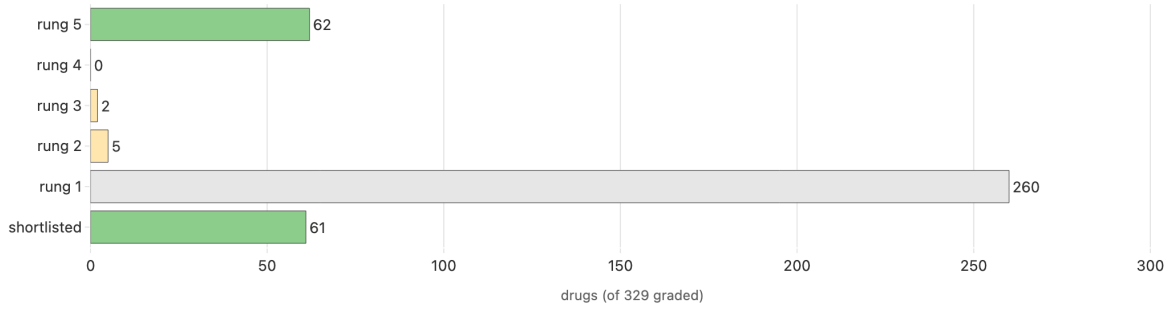
> **Legend** — the number is the rung; **green** a curated result and rung 4–5; **yellow** a curated result at rung 2–3; **blue** the automated tests alone (already being tried in the tumour — not a finding of this work); **red** set aside (a spindle poison or the standard chemotherapy); `*` inherited (proxy) evidence. Drugs with a curated result are listed first.

6.1 The eleven drugs with a cited mechanism result

#	drug (approved for)	direction	rung	the cited result	inherited?	given to children?
1	**lovastatin** (cholesterol)	D4 tumour	5	lovastatin itself slowed rhabdomyosarcoma cells; the tumour depends on making its own cholesterol and cannot take it up instead (Gizaw 2025, PMID 41249736). A completed phase-1 study gave a statin with chemotherapy to children with relapsed solid tumours including rhabdomyosarcoma (NCT02390843)	no	label section 8.4; studies admitting children
2	**simvastatin** (cholesterol)	D4	5	same class result; it is the statin in NCT02390843	yes (lovastatin)	studies admitting children
3	**nicotinamide** (vitamin B3)	D3 more BubR1	3	the NAD^+ precursor NMN raised BubR1 in mice, and more SIRT2 (the enzyme that stabilises BubR1) extended the life of BubR1-deficient mice (North 2014, PMID 24825348)	yes (NMN)	studies admitting children
4	**niacin** (cholesterol; vitamin B3)	D3	3	same result; niacin is the prescription-strength NAD^+ precursor	yes (NMN)	studies admitting children
5	**chloroquine** (malaria)	D2 stress	2	chloroquine itself preferentially killed cells with an extra chromosome in the reference screen (Tang & Amon 2011, PMID 21315436)	**no**	studies admitting children
6	**bortezomib** (myeloma)	D2	2	aneuploid human cells depend on degrading proteins faster and are more sensitive to proteasome inhibitors; myeloma patients' aneuploidy tracked their response (Ippolito 2024, PMID 39247952; Chui 2019, PMID 31530568)	yes (the drug class)	label 8.4; studies admitting children
7	**valproic acid** (epilepsy)	D3	2	HDAC2/3 strip the acetyl mark that protects BubR1 from being degraded (Park 2017, PMID 28985013; Choi 2009, PMID 19407811); an HDAC inhibitor would be expected to slow that loss — inferred, not shown	yes (class mechanism)	label 8.4; studies admitting children
8	**metformin** (diabetes)	D2	2	the AMPK activator AICAR was aneuploidy-selective in the same screen; metformin is the approved AMPK activator	yes (AICAR)	label 8.4; studies admitting children
9	**vorinostat** (lymphoma)	D3	2	same class mechanism as valproic acid	yes	label 8.4; studies admitting children
10	**hydroxychloroquine** (lupus, arthritis)	D2	2	chloroquine's result, inherited by its clinical analogue	yes (chloroquine)	label 8.4; studies

#	drug (approved for)	direction	rung	the cited result	inherited?	given to children?
						admitting children
11	**atorvastatin** (cholesterol)	D4	2	same class result as lovastatin; carries a paediatric indication on its label	yes (lovastatin)	label 8.4; studies admitting children

Reading the table honestly: **one** drug has a molecule-level animal result on the checkpoint itself — none. The deepest mechanistic lead (D3, more BubR1) rests on one mouse paper and is inherited by two vitamin-B3 forms. The best-supported *tumour* lead (statins) has a completed paediatric trial behind it and a molecule-level cell result. The only molecule-level *aneuploidy* result is chloroquine's.



> **Legend** — drugs counted once at their best rung; **green** rung 4–5 and the shortlist, **yellow** rungs 2–3, **grey** rung 1 (named, never submitted). Most of the 62 at rung 5 are there by the automated test alone: approved, trialled in rhabdomyosarcoma, used in children.

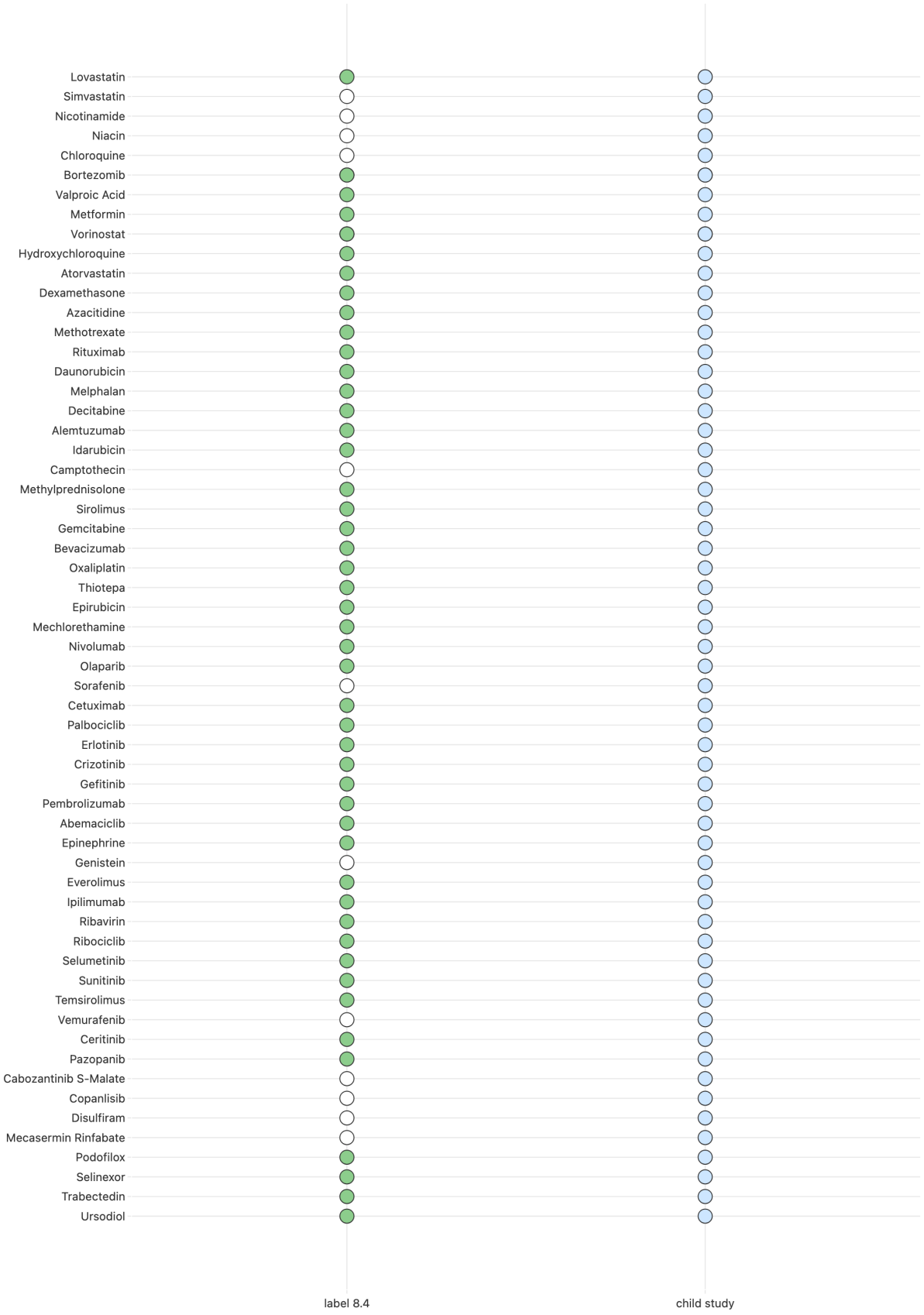
6.2 Fifty drugs already being tried in rhabdomyosarcoma — listed, not claimed

Fifty more approved drugs reach rung 5 automatically: they have a registered rhabdomyosarcoma trial and a paediatric pointer (dexamethasone, azacitidine, methotrexate, bevacizumab, the CDK4/6 and mTOR inhibitors, the checkpoint antibodies…). They are in `shortlist.tsv` with the verdict *context trial only – not a mechanism finding*, because the decoy control showed the same list comes out for a gene that has nothing to do with the checkpoint. They are the tumour's trial landscape, and a reader deserves to see them; they are not this work's finding.

6.3 The do-not table

verdict	drugs	why
caution — spindle poisons (13)	vincristine, vinorelbine, vinblastine, vinflunine, paclitaxel, docetaxel, cabazitaxel, eribulin, ixabepilone, epothilone D, colchicine, brentuximab vedotin, polatuzumab vedotin	these drugs work by jamming the division machinery and *relying on the checkpoint to hold the cell*; MVA cells have a weakened checkpoint and slip through (Suijkerbuijk 2010, PMID 20516114; Sinha 2019, PMID 30601084). Vincristine and vinorelbine are standard rhabdomyosarcoma chemotherapy: **a hypothesis for the treating team to weigh, never advice to withhold treatment**
do not — further checkpoint weakening (0 approved)	Mps1/TTK and Aurora inhibitors	aneuploid cancer cells are selectively killed by weakening the checkpoint further (Cohen-Sharir 2021, PMID 33505028) — the same logic makes a child whose checkpoint is constitutionally weak the worst possible candidate. None is approved; every one in the candidate table is phase 3 or earlier
standard of care (10)	vincristine, dactinomycin, cyclophosphamide, ifosfamide, doxorubicin, etoposide, irinotecan, temozolomide, carboplatin, cisplatin, topotecan	already the treatment for the tumour; giving them is not a repurposing. Named so a reader sees they were considered, not missed

6.4 Has each candidate been given to children?



> **Legend** — **green** a US label with a *Pediatric Use* section (8.4); **blue** at least one registered study admitting children, in any indication; empty = neither found. Every argued drug has at least one pointer. **A pointer is something to read, never a licence**: a label section can say "safety not established", and a study that admits children may have enrolled none.

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7. What this could change — for this child, for others with MVA, for diagnosis

> **Rubric:** Potential Impact — what independent validation of this work would be worth, said separately for the child in front of us, for the disease, and for recognising the condition.

7.1 For this child: one question a treating team can act on, and one experiment that was not run

The do-not table (§6.3) is not a list; it is a question with a name on it. Thirteen approved drugs are **spindle poisons** (they work by jamming the machine that pulls chromosomes apart, and then relying on the checkpoint to hold the cell until the damage is noticed). This child's checkpoint is the broken part. Two of the thirteen — **vincristine** and **vinorelbine** — are standard rhabdomyosarcoma chemotherapy, so the question is not hypothetical for him.

the question a treating team could put	what this report supplies	what it does not supply
does a constitutionally weakened checkpoint change how this child's cells respond to vincristine or vinorelbine?	the mechanism (§4), the two published results the caution rests on (Suijkerbuijk 2010, PMID 20516114; Sinha 2019, PMID 30601084), and the full list of thirteen (§6.3)	any measurement in this child's cells, any dose implication, and any reason to withhold standard care. **This report flags the question; it does not answer it.**
is there an approved drug that reinforces the checkpoint instead?	the answer is **no** — direction D1 returned none, and every checkpoint drug on the market or in trials *weakens* it (§2, §6.3)	a reason to stop looking: the finding is about today's approved pharmacopoeia, not about what is achievable

The one experiment that would move the deepest lead. D3 (raise the amount of working BubR1) is the only direction that touches the broken protein itself, and it sits at rung 3 on a single mouse study in which the NAD^+ precursor **NMN** — not an approved drug — raised BubR1; niacin and nicotinamide, which *are* approved, inherit that result as a `proxy` (North 2014, PMID 24825348). The experiment that would turn a proxy animal result into a molecule-level one: expose cells carrying a *null + weakened* `BUB1B` genotype to **nicotinamide or niacin themselves**, and measure BubR1 protein level and checkpoint strength against an untreated control. **This is a proposal. It was not run, and nothing in this report is evidence of what it would show.**

7.2 For others living with MVA: two results that belong to the mechanism, not to this gene

MVA is not a one-gene disease, and its other known causes sit in the same chromosome-segregation machinery: `BUB1B` is the core of the checkpoint itself, `TRIP13` resets the MAD2 switch so the checkpoint

signal can be rebuilt, and `CEP57` is part of the centrosome that builds the spindle the checkpoint watches. Whichever gene is broken, the cell ends up with **a leaky brake**.

Two of this report's results are therefore arguments about a weakened checkpoint rather than about this child's particular gene:

1. **No approved drug reinforces the checkpoint** (direction D1, §2). Every marketed or trialled checkpoint drug is built to *weaken* it, because weakening it is what kills cancer cells. A child whose checkpoint is constitutionally weak is on the wrong side of that entire development programme — and that is true for any MVA genotype, not only this one.
2. **The spindle-poison caution** (§6.3). It rests on the checkpoint being leaky, which is the shared endpoint of every MVA gene above.

Said plainly: this is reasoning *from* the mechanism, not a measurement across genes. The claim that the same pipeline reproduces both results on a different MVA gene is registered as **H066**, and its test **post-dates this draft** — see §9 and `../iteration-2/report.md`.

7.3 For diagnosis: a pipeline and its stated limits, not a measurement

Be precise about what *this draft* contributes to recognising the condition. It contributes the **pipeline and the limits it prints**, not a number:

contribution	what it is	what it is not
the mechanism card as a hand-off format	a structured record — gene, allele *classes* (never positions), pathway, tumour context, the measured facts and the unmeasured ones — that a downstream tool consumes without a person retyping the case	a diagnostic test
an explicit *unmeasured* row	this draft's card (version 3) records the aneuploid fraction as **unmeasured**, and says why: the karyotype statistic run in iteration 1 has no power for mosaic gains (§4.1)	a measurement. A pipeline that prints "I cannot see this" helps diagnosis only by stopping a reader trusting a number nobody took
the burden itself	**not in this draft.** The powered analysis was run afterwards and iteration 2 reports fact F8 as measured — `../iteration-2/report.md`	anything this report should be cited for

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8. What is new here, against standard practice

> **Rubric:** Innovation — what this pipeline does that a repurposing write-up does not usually do, and, just as importantly, what it does not claim.

The prior art, named accurately. Computational drug repurposing has mature tooling, much of it graph-based: **PrimeKG** (a biomedical knowledge graph, PMID 36732524) and **TxGNN** (a graph neural network scoring drugs against a disease, PMID 39322717) both take a **disease node** as input and return a ranked drug list. That is a different question from the one asked here, which starts from **one child's two alleles and the mechanism they imply**. The repo's own notes on both tools live in `docs/iteration4/cards/` and **post-date this draft**; they are named here as prior art, not as a result of this work.

what a repurposing report usually does	what this one does
a person hand-types the target and the literature queries, and the reader never sees which queries were tried	**one structured mechanism card** — gene, allele classes, pathway, tumour context, four directions with seed genes — drives every query; swap the card, get another case (§5.1, §11)
returns a ranked list, with the ranking function's weights implicit	a **five-rung evidence ladder written down before the search**, with the automated test and the person-read test **printed side by side** on every row, and inherited class-level evidence flagged `proxy` (§5.2)
no negative control: the pipeline is never run on an input that should return nothing	the **same pipeline re-run on a decoy gene** — the child's clinical side kept, the gene swapped, twice — with the overlap measured against a threshold fixed in advance, and the one direction that failed it argued differently as a result (§5.3). *To our knowledge* this is the part most repurposing work has no equivalent for
candidates only	a **do-not table as a first-class output**: 13 spindle poisons, the checkpoint inhibitors, and the standard of care, named so a reader can see they were considered rather than missed (§6.3)
confidence asserted after the fact, in prose	an **append-only claim register**: each claim carries its prediction *and* its refutation condition, filed before the test, and a correction stays visible instead of being edited away (§12)

What is **not** claimed: no new model, no new algorithm, no new biology. The novelty is in the process — pre-registration, a negative control, and never silently merging machine-found with person-read evidence. "To our knowledge" is doing real work in the table above: no systematic survey of repurposing methods was run, so read it as *we are not aware of a comparable decoy control*, never as *none exists*.

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9. Could this run for the next child?

> **Rubric:** Scalability — what it takes to point this pipeline at another person, another gene or another disease, and what breaks first when you do.

Case-agnostic by construction. The pipeline has one input — the mechanism card — and five commands (§11). Nothing about this child is compiled into a module: the gene, the allele classes, the pathway, the disease, the tumour context and the four directions are all fields on the card, and every table written out carries a provenance manifest naming its queries and its retrieval time. The decoy runs in §5.3 are the demonstration: the same code, one field changed, no edit.

The cost shape, measured on this case.

resource	this case
compute	a laptop. Two runs of about ten minutes each — the snapshot and the enrichment — most of it polite waiting between requests; **seconds when the replies are cached**. No cloud, no GPU, no cluster
data	seven public APIs; no licence, no patient-scale file to move
a person	the curated rungs: thirteen abstracts, read on one day

What breaks at scale. Three things, in the order they bite:

1. **The politeness interval on public APIs.** The runtime is per-drug and dominated by waiting between requests, not by computation. Ten minutes for one card is fine; a thousand cards is not the same run

repeated, it is a rate-limit and caching problem this pipeline does not solve.
2. **The curated rungs are the human bottleneck.** Rungs 2 and 3 exist only because a person read a paper — here, thirteen abstracts, one reader, one day (§10, item 5). The automated rungs scale freely; the rungs that carry the actual mechanism argument do not.
3. **One drug database was never queried.** DrugBank needs a licence for programmatic access, so it is absent from every count in this report (§10, item 6). At scale that is a licensing decision, not an engineering one.

A transfer onto a genuinely different case exists — and it is not this draft's. The pipeline was run on a card for a different MVA gene, with its own disease identifiers, its own clinical terms, its own tumour context and no curated evidence. That run's tables are stamped 2026-09-19, **after this draft was written on 2026-09-18**; its claim is registered as H066 and its results belong to iteration 2: `../iteration-2/report.md`. **No number from that run appears in this report**, because this report is not written from it.

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10. What this report cannot tell you

> **Rubric:** Scientific Rigor — a report a panel can check is one that names its own limits before the panel has to find them.

1. **Whether any of these drugs helps this child.** Rung 3 is one mouse paper; rung 2 is a dish. The challenge's words hold: hypotheses for follow-up.
2. **How aneuploid this child is.** The delivered data was tested with a statistic that has no power for mosaic gains; the powered analysis is planned (iteration 2, step 8) and has not run. Direction D2 inherits this gap entirely.
3. **Whether the weakened BubR1 copy is unstable or inactive.** It sits outside every catalogued domain; the structural model is confident about the region but says nothing about function. D3 is therefore argued as *more BubR1*, not *repair this BubR1*.
4. **Anything about a drug the public sources never listed** beyond what a person put in the curated table. Four of the eleven argued drugs (chloroquine, hydroxychloroquine, niacin, nicotinamide) are there only because a person read the paper; their rows say `curated_only`.
5. **The curated rungs are one person's reading of thirteen abstracts on one day.** Every row carries its PubMed id so the reading can be checked.
6. **DrugBank was not queried** (it needs a licence for programmatic access); Open Targets, ChEMBL, DGIdb, openFDA, Reactome, ClinicalTrials.gov and PubMed were.

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11. Reproducibility

> **Rubric:** Scalability and Scientific Rigor — the five commands that rebuild every number in this report, and the cost of running them.

Every step is one command, case-agnostic (swap the card, get another case), with a manifest:

```
uv run mva-drugs snapshot                          # card → 5 tables, 7 public APIs, cached repl
uv run python tools/track2_decoys.py ...           # a decoy card and its snapshot (×2)
uv run python tools/track2_enrich.py               # literature, paediatric trials, labels, ladde
uv run python tools/render_track2_figures.py       # the figures in this report
uv run python tools/record_track2_lineage.py       # 63 lineage edges into the warehouse ledger
```

Runtime on a laptop: the snapshot ~10 minutes cold (mostly polite waiting between requests), the enrichment ~10 minutes cold, both seconds when cached. Code: the public repository named in the submission form; every table is under `ressources/track2/`, every entry under `docs/iteration2/step-7/`. **No patient data appears anywhere in Track 2** — gene symbols, drug names, trial ids and PubMed ids only; the mechanism card names allele *classes*, never positions.

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12. Where every claim lives

> **Rubric:** Scientific Rigor — each claim, its register id, its verdict and the entry that tested it, so a reader can audit the reasoning rather than trust it.

claim	register id	verdict · confidence	entry
D1 yields no submittable drug and becomes the do-not table	H045	supported · 0.80	113
D2 holds an approved compound with a molecule-level cell result	H046	supported · 0.70	113
D3 has an approved route on the NAD^+/SIRT2 axis with animal evidence, by proxy	H047	supported · 0.60	113
D4 holds a non-chemotherapy approved drug at rung ≥ 2 with paediatric data	H048	supported · 0.80	113
the automated stages are non-specific (a direction overlaps a decoy ≥ 0.10)	H049	supported · 0.85	112
every argued drug has a paediatric pointer	H050	supported · 0.70	113
no approved drug acts directly on BubR1	H011 (iteration 1)	supported · 0.80	078

Confidence is a transparent triage heuristic printed with its arithmetic, not a calibrated probability; the register is append-only and shows every revision, including one committed too early and corrected in the open (entry 113).